MIL-L-11566C(ME)
1 September 1978
SUPERSEDING
MIL-L-11566B(ME)
4 August 1972

MILITARY SPECIFICATION

LEVEL, LEVEL ROD

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of level which will indicate the verticality of level rod and range pole placement.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-3534	- Surveying Instruments and Accessories, Packaging of.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:20Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking for US Military Property.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The level shall consist of a circular level vial adjustably mounted on a right-angle support, as shown on Figure 1 (see 6.3), and as specified herein.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.2.1 Metal. Metal shall be corrosion-resistant material such as magnesium, aluminum, brass, or corrosion-resistant steel.

3.3 Environmental conditions.

3.3.1 Operating temperature. The level shall perform in any ambient temperature from plus 110° F (43° C) with full impact of solar radiation of 360 Btu/sq ft/hr (1135 watts per square meter) for not less than 4 hours to minus 25° F (minus 32° C).

3.3.2 Storage temperature. The level shall not be damaged by a storage temperature of plus 160° F (71° C) for a period of 1 hour or minus 30° F (minus 34° C) for a period of 6 hours with no solar radiation.

Source: https://assist.dla.mil -- Downloaded: 202640-17T18:20Z
Check the source to verify that this is the currenversion before use.

3.4 Circular level vial. The circular level vial shall have a sensitivity of 20 minutes, plus or minus 10 percent, of arc for 2 millimeter displacement. The reference circle shall have a diameter of 8 millimeters, plus or minus 0.5 millimeter. The bubble shall have a diameter of 7 millimeters, plus or minus 0.5 millimeter at 75° F (24° C). The mounted vial shall be adjustable on the bracket of the support at three points. The vial shall be firmly positioned within the mounting.

3.5 Identification marking. Each level shall be identified in accordance with MIL-STD-130.

3.6 Finish and painting. Unless otherwise specified (see 6.2), finish and painting shall be at the option of the contractor.

3.7 Workmanship. The surfaces of the level shall be smooth and shall be free from rough edges and burrs.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:20Z
Check the source to verify that this is the current version before use.

4.3.2 Examination.

4.3.2.1 Samples. Samples selected in accordance with 4.3.1 shall be examined for the major and minor defects specified in 4.4.1. AQL shall be 1.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.3.3 Tests.

4.3.3.1 Samples. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 1.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The level shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Mounting not as specified.
103. Circular vial not adjustable as specified.
104. Marking missing, incomplete, or illegible.

Minor

201. Workmanship not as specified.

4.4.2 Tests.

4.4.2.1 Environmental conditions.

4.4.2.1.1 High temperature. The level shall be tested as specified in MIL-STD-810, Procedure II, Method 501.1, except that Step 6 of Procedure II shall be omitted. The maximum ambient operating temperature shall be 110° F (43° C) with full impact of solar radiation for a period of not less than 4 hours. The maximum ambient storage temperature shall be 160° F (71° C) for a period of 1 hour. Nonconformance to 3.3 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:20Z
Check the source to verify that this is the current version before use.

4.4.2.1.2 Low temperature. The level shall be tested as specified in MIL-STD-810, Procedure I, Method 502.1. The minimum ambient storage temperature shall be minus 30° F (minus 34° C) with no solar radiation for a period of not less than 6 hours. The minimum ambient operating temperature shall be minus 25° F (minus 32° C). Nonconformance to 3.3 shall constitute failure of this test.

4.4.2.2 Vial sensitivity tests. Place the enclosed vial on a leveled hinged beam, one end of which is moved by means of a micrometer screw, each revolution of which results in a known angular movement of the beam. With a scale, measure the bubble displacement after moving the beam through an arc of 20 minutes plus or minus 10 percent. Nonconformance to 3.4 shall constitute failure of this test.

4.5 Inspection of packaging. The preservation, packing, and marking for Level A or Level B shall be examined to determine compliance with the Quality Assurance Provisions of MIL-S-3534. The preservation, packing, and marking for Commercial shall be examined to determine compliance with MIL-STD-1188.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each level shall be preserved in accordance with the Level A preservation and packaging requirements of MIL-S-3534.

5.1.2 Commercial. Lach level shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Levels, preserved as specified in 5.1, shall be packed in accordance with the Level A packing requirements of MIL-S-3534.

5.2.2 Level B. Levels, preserved as specified in 5.1, shall be packed in accordance with the Level B packing requirements of MIL-S-3534.

5.2.3 Commercial. Levels, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

Source: https://assist.dla.mil -- Downloaded: 202640-17T18:20Z
Check the source to verify that this is the currenversion before use.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for Commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The level is intended for use as a plumbing device for leveling rods and range poles used in surveying.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When finish and painting is other than the contractor's option (see 3.6).
(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 Information figures. Figure 1 shows a type of level which has been found acceptable; however, the figure is included for illustration only and is not intended to preclude the furnishing of another level which conforms to this specification.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

Custodian:
Army - ME

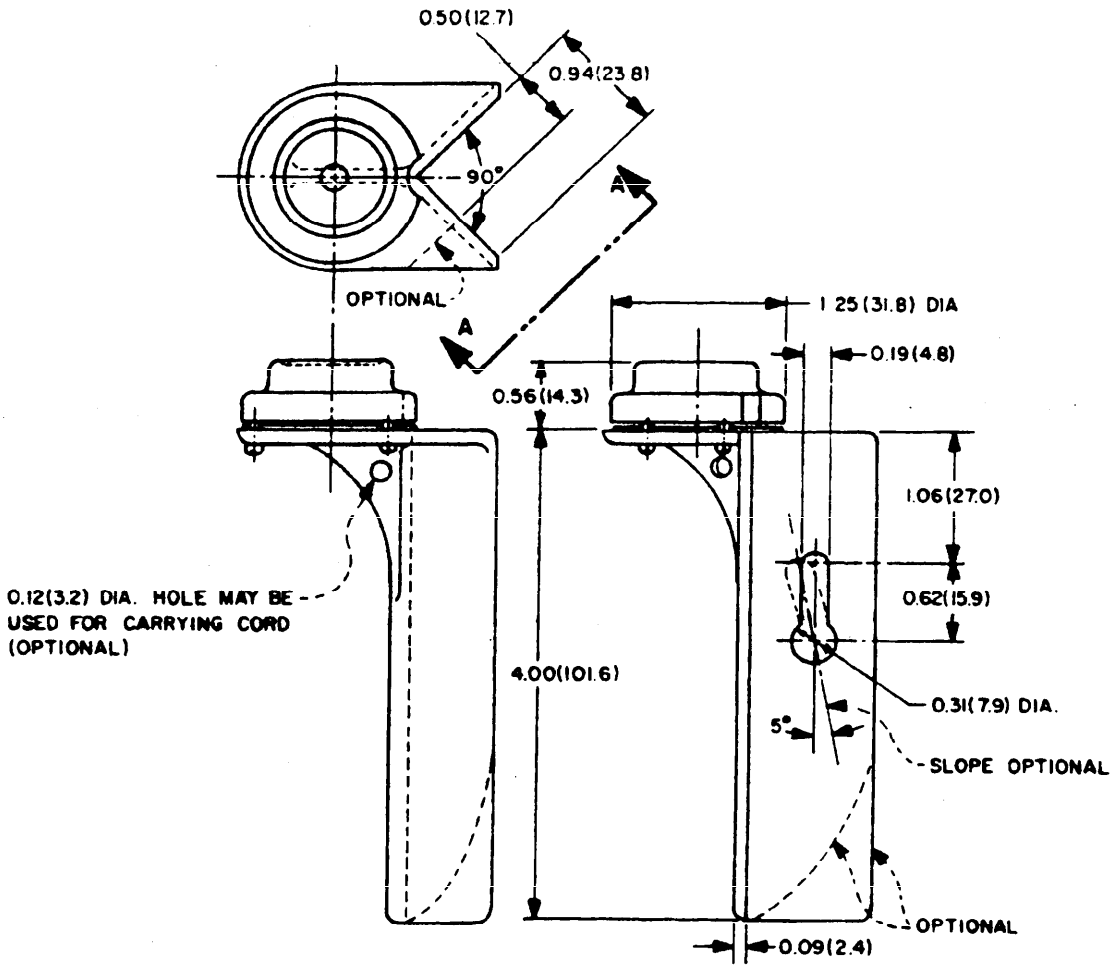
Review activity:
DLA - GS

User activity:
Army - CE

Preparing activity:
Army - ME

Project 6675-A217

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:20Z
Check the source to verify that this is the current version before use.



☆U.S. GOVERNMENT PRINTING OFFICE: 1978 — 703-122/4397

FIGURE 1. Level, level rod.

X-3377

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:20Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:20Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22 R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-L-11566C(ME) Level, Level Rod

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 202640-17T18:20Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Mobility Equipment Research
and Development Command
ATTN: DRDME-DS
Ft. Belvoir, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:20Z
Check the source to verify that this is the current version before use.